Joint Rate and Coverage Design for UAV-Enabled Wireless Networks with Underlaid D2D Communications

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***Abstract*—Unmanned aerial vehicle (UAV) can be utilized as aerial base station to provide emergency coverage in disasters, due to its flexible deployment, potential mobility and low cost. However, the wireless coverage of a UAV is limited because of the constrained transmit power. Therefore, device-to-device (D2D) transmission can be exploited to extend wireless coverage of UAV. In this paper, we aim to jointly maximize the weighted sum rate of the UAV-served users and the total number of users with wireless service for UAV-enabled wireless networks with underlaid D2D communications. The optimal set of UAV-served users are determined, while satisfying the information causality constraint. The formulated problem involving binary variables is non-deterministic polynomial (NP)-hard and extremely difficult to solve. Accordingly, we propose the algorithm to transform the problem into solving linear programming problem with low-complexity, and obtain a suboptimal solution. Simulation results demonstrate that the proposed joint rate and coverage design achieves better system performance.**

Keywords-unmanned aerial vehicle (UAV), device-to-device (D2D), wireless coverage, sum rate, linear programming problem.

I. INTRODUCTION

UAVs can serve as flying base stations to provide emergency communications in disaster regions where the cellular infrastructure has been damaged [1]. Compared with conventional static base stations, UAV-enabled base stations can be deployed more swiftly and improve system performance by dynamically adjusting locations due to the fact that the channel between the aerial UAV and ground user is nearly dominated by line-of-sight (LoS) in most scenarios [2], [3].

Thus, UAV-enabled wireless networks are attracting increasing interests from both industry and academic. In [4], three typical cases of UAV-aided wireless communications are summarized by Zeng *et al*., including UAV-aided ubiquitous coverage, UAV-aided relaying, and UAV-aided information dissemination. In [5], Christy *et al*. design UAV trajectories to adapt to disaster condition and satisfy the energy constraint simultaneously.

However, a single UAV cannot connect large amount of users due to the constrained transmit power. In addition, it is difficult to deploy many UAVs to provide wide wireless coverage simultaneously because the rescue resource is extremely scarce in disastrous areas. There has been considerable recent studies on device-to-device (D2D) communications between wireless users over the licensed spectrum [6]. Therefore, to address this challenge, D2D connections reusing existing licensed spectrum resource [7] can be established to extend the coverage of UAV effectively, which allow direct transmission between two nearby users [8]-[10].

In [11], Cheng *et al*. study the user association for a dual-UAV enabled wireless network with D2D communications. The formulated problem is a combinatorial optimization problem involving binary variables, the authors propose a learning-based clustering algorithm by viewing it as a clustering problem. In [12], Liu *et al*. propose multi-antenna transceiver design and multi-hop D2D communication to guarantee the reliable transmission and extend the UAV coverage for Internet of Things (IoT) in disasters. In [13], with the aim of maximizing the rate of a D2D pair for a downlink UAV-aided wireless communication system, the authors jointly optimize the transmit power of the UAV and D2D users, the flying altitude and location of the UAV and ground terminals allocated bandwidth. An iterative algorithm with low complexity is proposed to solve the formulated problem.

In this paper, we consider a scene where UAV and D2D connections are jointly leveraged to provide wireless service for more users when the communication infrastructures are damaged in disasters. To obtain better system performance, we propose a joint sum rate and coverage design under the information causality constraint. The formulated optimization problem is NP-hard and extremely complex due to involving binary variables. Accordingly, we propose an algorithm to obtain a suboptimal solution to the optimization problem by transforming it into solving linear programming problem, which can be solved by using existing optimization tools [14], such as CVX.

The rest of this paper is organized as follows. In Section II, system model for UAV-enabled wireless networks with underlaid D2D communications is described. Problem formulation is presented in Section III. Simulation results and discussion are shown in Section IV, followed by conclusions in Section V.

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II. SYSTEM MODEL

Consider a circular area in which a number of wireless users are assumed to be randomly and uniformly distributed. In the area, as shown in Fig. 1, the UAV placed at the area center is deployed to act as a flying based station and to serve a subset of those users. Users in the set can be either selected to download information from the UAV, namely UAV-served users, or obtain information from nearby users by forming D2D pairs. For convenience, Table 1 provides the major notations used in this paper.

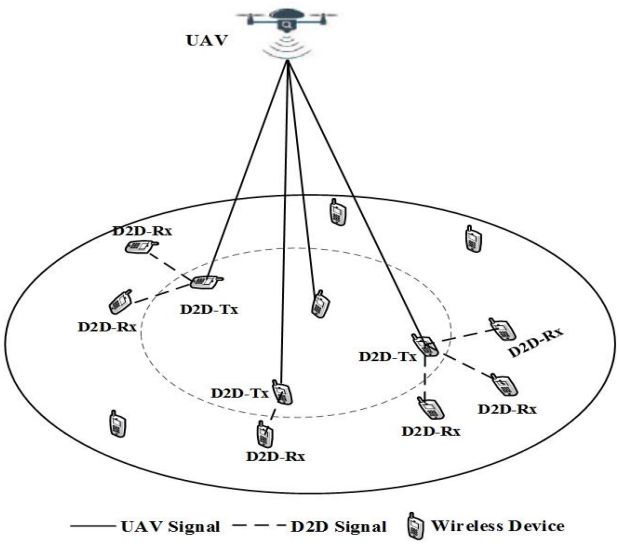

Figure 1. Investigated scenario of UAV-enabled networks with underlaid D2D communications

A. Air-to-Ground Channel Link

The UAV is located at the center of the circular area with height H. The set of ground users is denoted by $\mathcal{N}=\{1,2,...n,...,N\}$. The channel between the UAV and each UAV-served user is dominated by the LoS path. Denote (x_n, y_n) as the horizontal location of the nth user, and binary variables $\{i_n\}$ as the indicator of the UAV-to-user connection. If $i_n=1$, the nth user is served by the UAV, otherwise $i_n=0$. $\mathcal{M}=\{1,2,...,m,...M\}$ is the set that contains all probable UAV-served users, which yields the constraint as

$$\sum_{n=1}^{N} i_n \le M. \tag{1}$$

The channel fading coefficient g between UAV-served user and UAV is random or time-varying. Therefore, we replace it by β, the average value of instantaneous fading power $|g^2|$, as

$$\beta=\mathbb{E}\left(|g|^2\right). \tag{2}$$

Orthogonal frequency division multiple access (OFDMA) is designed for multiple UAV-served users. Therefore, the interference between UAV-served users is not considered. The received signal-to-noise ratio (SNR) of the nth UAV-served user at distance $d_0=1m$ can be expressed as

TABLE I. MAJOR NOTATIONS

Notation	Definition
$\mathcal{N}$	User set
N	Number of users
$\mathcal{M}$	UAV-served user set
M	Number of UAV-served users
x_n	x coordinate of the nth user
y_n	y coordinate of the nth user
i_n	Indicator of the UAV-to-user connection
j_{nm}	Indicator of the D2D connection
N_n	Number of users served by the nth UAV-served user via D2D connections
α	Reference path loss at distance $d_0=1m$
g	Channel fading coefficient
β	Average value of instantaneous fading power
γ_n	Signal-to-noise ratio of the nth UAV-served user
σ^2	Noise power
p_{UAV}	UAV transmission power
H	UAV altitude
d_{D2D}	Maximum distance of D2D connection
d_{nm}	Distance between the mth UAV-served user and the nth D2D receiver
R_n	Channel rate of the nth user and UAV
ω	Weight

$$\gamma_n=\frac{\frac{p_{UAV}\alpha\beta}{\sigma^2}}{H^2+x_n^{\ 2}+y_n^{\ 2}},\ \forall n\in\mathcal{N}, \tag{3}$$

where p_{UAV} is the transmit power of UAV, and α is the reference path loss at distance $d_0=1m$.

Furthermore, the channel rate of the nth user and UAV is given by

$$R_n=i_n\log\left(1+\gamma_n\right),\ \forall n\in\mathcal{N}. \tag{4}$$

B. D2D Connections

The regular ground link can be modeled by the practical Rayleigh fading channel. To guarantee the quality of service (QoS) of D2D transmission, we assume that a D2D link can be established only when the distance between D2D transceiver and D2D receiver is less than d_{D2D}.

We define binary variables $\{j_{nm}\}$ as the indicator of the D2D connection. If the nth ground user connects to the mth UAV-served user via D2D connection, then $j_{nm}=1$;

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otherwise $j_{nm} = 0$. Furthermore, we have the following constraints.

All the users do not need to communicate with themselves, which yields the constraint as

$$j_{nn} = 0,\ \forall n \in \mathcal{N}. \tag{5}$$

Each user can only communicate with at most one transmitter, either the UAV or the UAV-served user, at one time, which follows the constraint

$$i_n + \sum_{m=1}^{N} j_{nm} \le 1,\ \forall n \in \mathcal{N}. \tag{6}$$

As shown in (6), if $i_n = 1$ and $\sum_{m=1}^{N} j_{nm} = 0$, the nth user will obtain content from the UAV. If $i_n = 0$ and $\sum_{m=1}^{N} j_{nm} = 1$, the nth user will obtain content from the mth UAV-served user via D2D connection. Also, we can define N_n as the number of users that can be served by the nth UAV-served user via D2D connections, which is

$$N_n = \sum_{m=1}^{N} j_{nm},\ n \in \mathcal{N}. \tag{7}$$

Furthermore, the information causality constraint is

$$j_{nm} \le i_m,\ \forall n, m \in \mathcal{N},\ n \ne m. \tag{8}$$

As shown in (8), the D2D transmitter can transmit information to its corresponding D2D receivers only after it has been served by the UAV, i.e., if $i_m = 1$, then $j_{nm} = 0$ or $j_{nm} = 1$. If $i_m = 0$, then $j_{nm} = 0$, and the nth user cannot obtain content either from the UAV or from the mth UAV-served user.

The distance between the mth UAV-served user, which is the D2D transceiver, and the nth D2D receiver can be denoted as

$$d_{nm} = \sqrt{(x_n - x_m)^2 + (y_n - y_m)^2}, \forall n, m \in \mathcal{N}. \tag{9}$$

To guarantee the QoS of the D2D transmission, we have the constrain of maximum distance of D2D connection. If the distance is more than d_{D2D}, the D2D connection cannot be established, which is

$$d_{nm} \le d_{D2D}, \forall n, m \in \mathcal{N}. \tag{10}$$

$$j_{nm} = 0,\ if\ d_{nm} \ge d_{D2D},\ \forall n, m \in \mathcal{N}. \tag{11}$$

III. Problem Formulation

To achieve better system performance, we jointly maximize the weighted sum rate of the UAV-served users and the total number of users with wireless service. The optimization problem can be formulated as follows.

$$\max_{i,j}\ w\sum_{n=1}^{N} i_n R_n + (1-w)\sum_{n=1}^{N} N_n \tag{12}$$

$$s.t.\ i_n \in \{0,1\}, j_{nm} \in \{0,1\}, \forall n, m \in \mathcal{N}, \tag{13}$$

$$\sum_{n=1}^{N} i_n \le M, \tag{14}$$

$$j_{nm} = 0,\ if\ n = m,\ \forall n, m \in \mathcal{N}, \tag{15}$$

$$i_n + \sum_{m=1}^{N} j_{nm} \le 1,\ \forall n \in \mathcal{N}, \tag{16}$$

$$j_{nm} \le i_m,\ \mathrm{n} \ne \mathrm{m},\ \forall n, m \in \mathcal{N}, \tag{17}$$

$$d_{nm} \le d_{D2D},\ else\ j_{nm} = 0,\ \forall n, m \in \mathcal{N}, \tag{18}$$

where $i = \{i_n, \forall n \in \mathcal{N}\}, j = \{j_{nm}, \forall n, m \in \mathcal{N}\}$, and $N_n = \sum_{m=1}^{N} j_{nm}, \forall n \in \mathcal{N}$. N_n is the number of users that the nth UAV-served user can communicate with via D2D connections. ω is the weight assigned to the sum rate of the UAV-served users, and 1-ω is the weight assigned to the total number of D2D-served users, which satisfies

$$0 \le \omega \le 1. \tag{19}$$

The optimization problem is NP-hard due to the fact that it involves binary variables i_n and j_n. Because of the extremely high computational complexity of NP-hard problem, it is difficult to obtain its optimal solution with tolerable computational complexity. To solve the problem approximately and obtain the suboptimal solution to it, we propose the algorithm to relax the binary variables i_n, j_n into continuous ones as

$$0 \le i_n \le 1,\ 0 \le j_{nm} \le 1,\ \forall n, m \in \mathcal{N}. \tag{20}$$

Therefore, the optimization problem can be reformulated as

$$\max_{i,j}\ w\sum_{n=1}^{N} i_n R_n + (1-w)\sum_{n=1}^{N}\ \sum_{m=1}^{N} j_{nm} \tag{21}$$

$$s.t.\ 0 \le i_n \le 1,\ 0 \le j_{nm} \le 1,\ \forall n, m \in \mathcal{N}, \tag{22}$$

$$\sum_{n=1}^{N} i_n \le M, \tag{23}$$

$$j_{nm} = 0,\ if\ n = m,\ \forall n, m \in \mathcal{N}, \tag{24}$$

$$i_n + \sum_{m=1}^{N} j_{nm} \le 1,\ \forall n \in \mathcal{N}, \tag{25}$$

$$j_{nm} \le i_m,\ \mathrm{n} \ne \mathrm{m},\ \forall n, m \in \mathcal{N}, \tag{26}$$

$$d_{nm} \le d_{D2D},\ else\ j_{nm} = 0,\ \forall n, m \in \mathcal{N}. \tag{27}$$

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This optimization problem is in linear programming (LP) standard form and thereby can be solved by using existing optimization tools, such as CVX.

IV. Simulation Results And Discussion

A. Simulation Setup

The target region is assumed to be a square area of size. The horizontal location of the UAV is assumed to be (0, 0). The altitude of the UAV is assumed to be $H = 100m$. Assume that N=100 users are randomly and uniformly distributed in a 2D area of $100 \times 100m^2$. The UAV can at most serve M=7 users directly. The list of key simulation parameters are summarized in Table 2.

B. Results and Discussions

When ω=1, the optimization problem is simplified to only maximize the sum rate of the UAV-served users. When $\omega = \frac{1}{3}$ or $\omega = \frac{2}{3}$, the weighted sum rate of the UAV-served users and the total number of users with wireless service can be jointly maximized. As shown in Fig. 2, the suboptimal solution to the proposed problem can be obtained by relaxing it into solving linear programming problem.

TABLE II. Simulation Parameters

Definitions	Parameters	Values
UAV altitude	H	100m
UAV transmission power	p_{UAV}	0.1w
Noise power	σ^2	100dBm
Signal-to-noise ratio	SNR	60dB
Path loss component	α	-60dB
Average value of the instantaneous fading power	β	1
Number of users	N	100
Number of UAV-served users	M	7
Maximum distance of D2D connection	d_{D2D}	10m
Weight	ω	$\frac{1}{3}, \frac{2}{3}, 1$

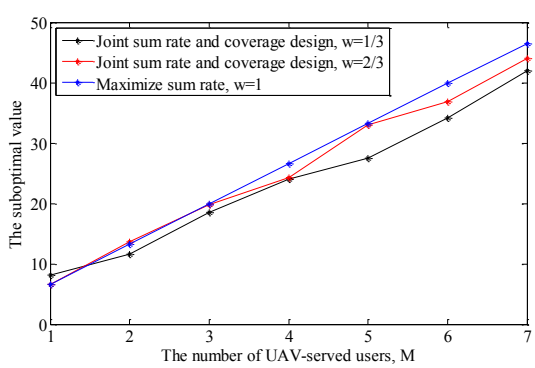

Figure 2. The suboptimal value with different number of UAV-served users

In Fig. 3 and Fig. 4, the UAV is marked by black ☆. The ground users are marked by blue ● and the UAV-served users are marked by red ◇.

When ω=1, the optimization problem is simplified to only maximize the sum rate of the UAV-served users. From Fig. 3, we can observe that users, near the UAV, with better air-to-ground link are selected to be served by UAV to maximize the sum rate. However, the number of users served by a single UAV is relatively limited.

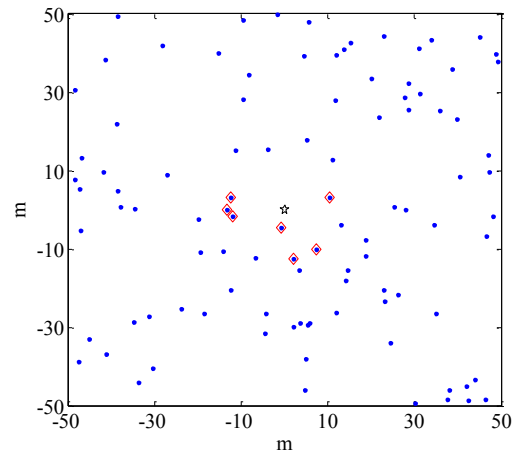

Figure 3. The distribution of UAV-served users when ω=1.

When $\omega = \frac{2}{3}$, the weighted sum rate of the UAV-served users and the total number of users with wireless service can be jointly maximized. The optimal set of UAV-served users are determined. As shown in Fig. 4, users that near the UAV and can serve more users via D2D connections are served by the UAV.

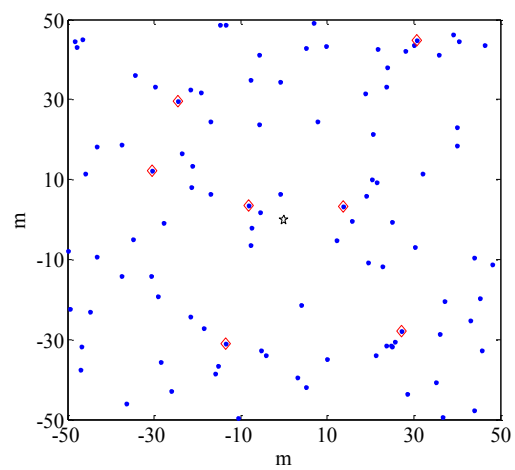

Figure 4. The distribution of UAV-served users when $\omega = \frac{2}{3}$.

In Fig. 5, we can observe that in the proposed joint sum rate and wireless coverage design, more users can obtain wireless services via D2D connections.

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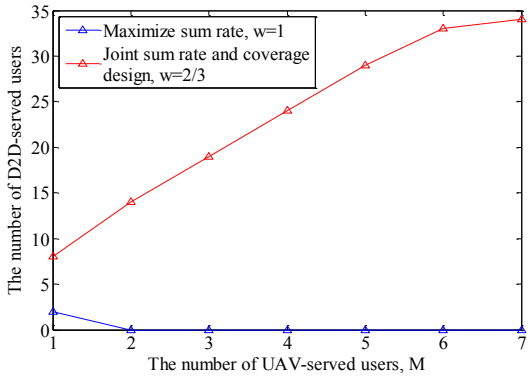

Figure 5. The total number of users served via D2D with different number of UAV-served users

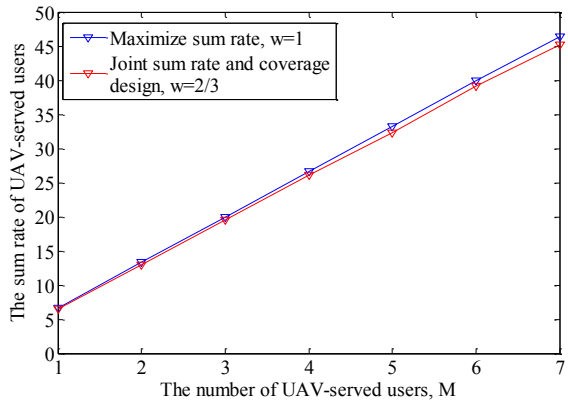

Figure 6. The sum rate of UAV-served users with different number of UAVserved users.

As shown in Fig. 6, the results of the two cases are similar. Although the proposed joint sum rate and wireless coverage design provided a slight decrease in the sum rate of UAV-served users, this can be tolerated by the fact that the proposed design still enhance the wireless coverage which is desirable for UAV-enabled wireless networks.

V. CONCLUSION

In this paper, we propose a joint rate and coverage design for the UAV-enabled wireless networks with underlaid D2D communications. To obtain a suboptimal solution to the formulated NP-hard problem, we propose the algorithm to relax the formulated problem into solving linear programming problem. Simulation results show that the proposed algorithm performs well. In the future work, other channel model cases will be investigated, such as considering the small-scale fading and non-line-of-sight (NLoS) probability.

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REFERENCES

[1] J. Lyu, Y. Zeng, R. Zhang, "Cyclical multiple access in UAV-aided communications: A throughput-delay tradeoff," IEEE Wireless Commun. Lett, vol. 5, pp. 600-603, 2016.

[2] J.Zhang, Y. Zeng, R. Zhang, "Spectrum and energy efficiency maximization in UAV-enabled mobile relaying," Proc. Int. Conf. Commun. (ICC), pp. 1-6, 2017.

[3] A.A. Khuwaja, Y. Chen, N. Zhao, M.S. Alouini, P. Dobbins, "A survey of channel modeling for UAV communications," IEEE Commun. Surv. Tuts, vol. 20, pp. 2804-2821, 2018.

[4] Y. Zeng, R. Zhang, T.J. Lim, "Wireless communications with unmanned aerial vehicles: Opportunities and challenges," IEEE Commun. Mag, vol. 54, pp. 36-42, 2016.

[5] E.Christy, R.P. Astuti, B. Syihabuddin, B. Narottama, O. Rhesa, F. Rachmawati, "Optimum UAV flying path for device-to-device communications in disaster area," Proc. ICSigSys 16, pp. 318-322, 2017.

[6] E. Yaacoub, O. Kubbar, "Energy-efficient device-to-device communications in LTE public safety networks," IEEE Global Telecommun. Conf, pp. 391-395, 2012.

[7] K. Doppler, M. Rinne, C. Wijting, C.B. Ribeiro, K. Hugl, "Device-todevice communication as an underlay to LTE-advanced networks," IEEE Commun. Mag, vol. 47, pp. 42-49, 2009.

[8] A. Asadi, Q. Wang, V. Mancuso, "A survey on device-to-device communication in cellular networks," IEEE Commun. Surveys Tuts, vol. 16, pp. 1801-1819, 2014.

[9] L. Wei, R. Hu, Y. Qian, G. Wu, "Enable device-to-device communications underlaying cellular networks: Challenges and research aspects," IEEE Commun. Mag, vol. 52, pp. 90-96, 2014.

[10] F. Jameel, Z. Hamid, F. Jabeen, S. Zeadally, M.A. Javed, "A survey of device-to-device communications: Research issues and challenges," IEEE Commun. Surveys Tuts, vol. 20, pp. 2133-2168, 2018.

[11] F. Cheng, D. Zou, J. Liu, J. Wang, N. Zhao, "Learning-based user association for dual-UAV enabled wireless networks with D2D connections," IEEE Access, vol. 4, pp. 1-11, 2016.

[12] X. Liu, Z. Li, N. Zhao, W. Meng, G. Gui, Y. Chen, F. Adachi, "Transceiver design and multi-hop D2D for UAV IoT coverage in disasters," IEEE Internet Things Journal, pp. 1-13, 2018.

[13] W. Huang, Z. Yang, C. Pan, L. Pei, M. Chen, M. Shikh-Bahaei, M. Elkashlan, A. Nallanathan, "Joint Power, Altitude, Location and Bandwidth Optimization for UAV with Underlaid D2D Communications," IEEE Wireless Communications Letters, pp. 1-4, 2018.

[14] S. Boyd, L. Vandenberghe, "Convex optimization," Cambridge University Press, 2004.

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